

Practice midterm solution

Problem 1. Consider generating a random variable X such that

$$\mathbb{P}(X = i) = \frac{e^{-\lambda} \lambda^i / i!}{\sum_{j=0}^k e^{-\lambda} \lambda^j / j!}, \quad i = 0, 1, 2, \dots, k.$$

(a) Derive the inverse transform method to generate X .

(b) Suppose you run the following algorithm to generate X : First sample from a Poisson distribution with rate λ , and if this value falls into $\{0, 1, \dots, k\}$, you output it; otherwise you return to the sampling step and repeat until you obtain an output. Is this an acceptance-rejection procedure? If it is, state precisely all the algorithmic configurations in this procedure.

Solution

(a) We have

$$\mathbb{P}(X = i + 1) = \frac{\lambda}{i + 1} \mathbb{P}(X = i)$$

The following procedure generates X :

Step 1. Generate $U \sim \text{Unif}(0, 1)$.

Step 2. $i = 0$; $q = \frac{e^{-\lambda}}{\sum_{j=0}^k e^{-\lambda} \lambda^j / j!}$; $F = q$.

Step 3. If $U \leq F$, $X = i$ and stop.

Step 4. $q = \frac{\lambda}{i+1} \cdot q$, $F = F + q$, and $i = i + 1$.

Step 5. Go to Step 3.

(b) The algorithm suggested in the problem is given as follows;

Step 1. Generate $Y \sim \text{Pois}(\lambda)$.

Step 2. If $Y \leq k$, let $X = Y$ (accept); else go to Step 1.

Now consider the candidate function $g(x)$ as a pmf of a Poisson random variable with parameter λ , and $f(x)$ as the given pmf

$$g(x) = e^{-\lambda} \lambda^x / x!, \quad x = 0, 1, 2, \dots$$

Let $C = \left(\sum_{j=0}^k e^{-\lambda} \lambda^j / j! \right)^{-1}$, be the scaling constant. Let us rewrite the $f(x)$.

$$f(x) = C e^{-\lambda} \lambda^x / x!, \quad x = 0, 1, 2, \dots, k$$

Consider the ratio $\frac{f(x)}{g(x)}$ for any $x = 0, 1, 2, \dots, k$,

$$\frac{f(x)}{g(x)} = \frac{C e^{-\lambda} \lambda^x / x!}{e^{-\lambda} \lambda^x / x!} = C,$$

and for $x > k$, $\frac{f(x)}{g(x)} = 0$.

Therefore, $\frac{f(x)}{Cg(x)} = 1$ if $x \leq k$, and $= 0$ if $x > k$.

As you can see the ratio test against a uniform random variable, $\frac{f(x)}{Cg(x)} \geq U$, $U \sim Unif[0, 1]$, does not actually depend on the value of x or the value of C . Since it is either 1 whenever Y is less than k and we accept in this case or 0 when Y is larger than k in which case we reject. Thus, the candidate algorithm indeed applies Acceptance-Rejection method with some simplifications in the procedure.

Problem 2. Let $\alpha, \beta > 0$. Give a method for generating a random variable having distribution function

$$F(x) = 1 - \exp(-\alpha x^\beta), 0 < x < \infty.$$

- (a) Derive an inverse transform method to generate X .
(b) Suppose you insist to use an acceptance-rejection method to generate X , using a proposal that has the same distribution function as above but with a new $\tilde{\beta}$ such that $\tilde{\beta} > \beta$, and that your computer has an established routine to generate from this proposal distribution directly. Does this acceptance-rejection method work? Explain.

Solution

- (a) $U = 1 - \exp(-\alpha x^\beta)$. Solving for X in terms of U gives;

$$1 - U = \exp(-\alpha x^\beta), \quad \log(1 - U) = -\alpha x^\beta$$

$$\frac{\log(1 - U)}{-\alpha} = x^\beta, \quad x = \left(\frac{\log(1 - U)}{-\alpha}\right)^{\frac{1}{\beta}}$$

Thus, we can generate the random variable X with the following algorithm

Step 1. Generate $U \sim Unif(0, 1)$.

Step 2. Return $X = \left(\frac{\log(1-U)}{-\alpha}\right)^{\frac{1}{\beta}}$

In fact you can also write U instead of $1 - U$ in Step 2 for a more elegant presentation.

- (b) Recall that to use Acceptance-Rejection method, we need to be able to find C such that

$$\frac{f(x)}{g(x)} \leq C \quad \forall x$$

Now plug in $f(x) = \frac{dF(x)}{dx} = \alpha \beta x^{\beta-1} \exp(-\alpha x^\beta)$ and $g(x) = \alpha \tilde{\beta} x^{\tilde{\beta}-1} \exp(-\alpha x^{\tilde{\beta}})$ with $\tilde{\beta} > \beta$. This pdf is defined only on non-negative half-line so we should have some C such that $\frac{f(x)}{g(x)} \leq C \quad \forall x \geq 0$. Consider large x , i.e. take limit of $\frac{f(x)}{g(x)}$

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \lim_{x \rightarrow \infty} \frac{\alpha \beta x^{\beta-1} \exp(-\alpha x^\beta)}{\alpha \tilde{\beta} x^{\tilde{\beta}-1} \exp(-\alpha x^{\tilde{\beta}})} = \lim_{x \rightarrow \infty} \frac{\beta}{\tilde{\beta}} x^{\beta-\tilde{\beta}} \exp(-\alpha x^\beta + \alpha x^{\tilde{\beta}})$$

However, we see that in the term in the exponent, because $\tilde{\beta} > \beta$, this limit will blow up to infinity since the exponential term increases very fast and dominates the polynomial term. Thus, we can never find a C such that $\frac{f(x)}{g(x)} \leq C \quad \forall x$ is satisfied, because there is always a very large x that will make $\frac{f(x)}{g(x)}$ greater than C . The calculations related to the limit are as follows;

$$= \lim_{x \rightarrow \infty} \frac{\beta}{\tilde{\beta}} x^{\beta-\tilde{\beta}} \exp(\alpha x^{\tilde{\beta}}(1 - x^{\beta-\tilde{\beta}}))$$

since $x \rightarrow \infty$, for large x we have, $x^{\beta-\tilde{\beta}} \leq 1/2$ thus we can write

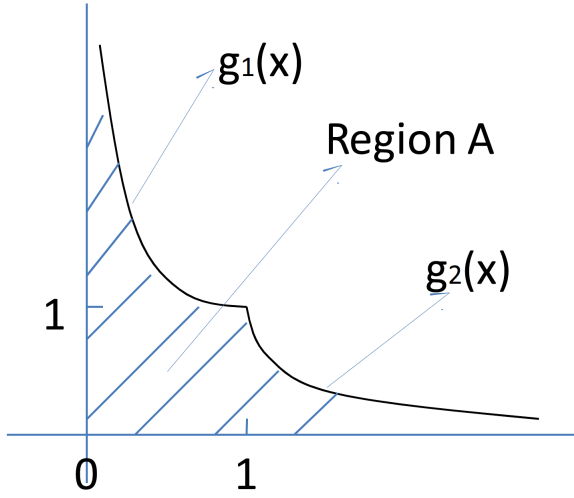
$$\geq \lim_{x \rightarrow \infty} \frac{\beta}{\tilde{\beta}} x^{\beta-\tilde{\beta}} \exp\left(\frac{\alpha}{2} x^{\tilde{\beta}}\right)$$

Let us also ignore constants and also divide by x^{β} which is a constant larger than 1. Then by classical calculus results, we get

$$\geq \lim_{x \rightarrow \infty} x^{-\tilde{\beta}} \exp\left(\frac{\alpha}{2} x^{\tilde{\beta}}\right) = \infty$$

Problem 3. Provide a method to sample uniform points in the region A under the curve $g_1(x)$ and $g_2(x)$ and the axis as indicated in the picture. Given $\alpha \in (0, 1)$, the curves $g_1(x)$ and $g_2(x)$ are described by

$$g_1(x) = x^{\alpha-1} \mathbf{1}\{0 < x \leq 1\}, \quad g_2(x) = x^{1/(\alpha-1)} \mathbf{1}\{1 \leq x < \infty\}.$$



Solution

The total area of region A is

$$\int_0^1 g_1(x) dx + \int_1^\infty g_2(x) dx = \int_0^1 x^{\alpha-1} dx + \int_1^\infty x^{\frac{1}{\alpha-1}} dx = \frac{2-\alpha}{\alpha}$$

Thus, the density function of (x, y) uniformly dropped in A is $\frac{\alpha}{2-\alpha} \mathbf{1}((x, y) \in A)$.

The marginal distribution of X is therefore:

For $x \in [0, 1]$

$$\mathbb{P}(X \leq x) = \frac{\alpha}{2-\alpha} \int_0^x t^{\alpha-1} dt = \frac{1}{2-\alpha} x^\alpha.$$

For $x \geq 1$

$$\mathbb{P}(X \leq x) = \frac{1}{2-\alpha} + \frac{\alpha}{2-\alpha} \int_1^x t^{\frac{1}{\alpha-1}} dt = 1 - \frac{1-\alpha}{2-\alpha} x^{\frac{\alpha}{\alpha-1}}.$$

Hence, the inverse of the cdf is given by

$$F^{-1}(y) = ((2 - \alpha)y)^{\frac{1}{\alpha}} \quad \text{if } y \in [0, \frac{1}{2 - \alpha}]$$

$$F^{-1}(y) = \left(\frac{2 - \alpha}{1 - \alpha} (1 - y) \right)^{\frac{\alpha - 1}{\alpha}} \quad \text{if } y \in [\frac{1}{2 - \alpha}, 1]$$

We can generate X using ITM and given X , Y is uniform $[0, g(X)]$.
The algorithm is given by

1. Generate $U_1 \sim Unif[0, 1]$.
2. If $U_1 \leq \frac{1}{2 - \alpha}$, $X = ((2 - \alpha)U_1)^{\frac{1}{\alpha}}$. Else, $X = \left(\frac{2 - \alpha}{1 - \alpha} (1 - U_1) \right)^{\frac{\alpha - 1}{\alpha}}$
3. Generate U_2 . Let $Y = g(X) \cdot U_2$ where $g(X) = g_1(X)$ if $X \in [0, 1]$ and $g(X) = g_2(X)$ if $X \geq 1$.
4. Return (X, Y) .

Problem 4. Consider the continuous nonnegative random variable X with probability density function

$$f_X(x) = \int_0^\infty \frac{9}{(1 + xt)^4} t^3 e^{-t^3} dt, \quad x \geq 0.$$

Derive an algorithm to generate X .

Solution We want to use the composition method but first let's do the following change of variables in the integral. $z = t^3$, $t = z^{\frac{1}{3}}$, $3t^2 dt = dz$. We get for $x \geq 0$.

$$f_X(x) = \int_0^\infty \frac{3z^{\frac{1}{3}}}{(1 + xz^{\frac{1}{3}})^4} e^{-z} dz = \int_0^\infty f_{Y_z}(x) f_Z(z) dz$$

where $f_Z(z) = e^{-z}$, i.e. Z is exponential of parameter 1. Now, we fix z . We have $f_{Y_z}(x) = \frac{3z^{\frac{1}{3}}}{(1 + xz^{\frac{1}{3}})^4}$.
The cdf of Y_z is given by

$$F_{Y_z}(t) = \int_0^t f_{Y_z}(x) dx = \int_0^t \frac{3z^{\frac{1}{3}}}{(1 + xz^{\frac{1}{3}})^4} dx = 1 - \frac{1}{(1 + tz^{\frac{1}{3}})^3}$$

The inverse of the cdf is given by

$$F_{Y_z}^{-1}(u) = \frac{1}{z^{\frac{1}{3}}} \left((1 - u)^{-\frac{1}{3}} - 1 \right).$$

We use ITM to generate Y_z The algorithm is given by

- Generate $Z = -\log(U)$ where $U \sim Unif[0, 1]$. Suppose $Z = z$.
- Generate $V \sim Unif[0, 1]$. Return $X = \frac{1}{z^{\frac{1}{3}}} \left((1 - V)^{-\frac{1}{3}} - 1 \right)$

Note: You may recognize $f_T(t) = 3t^2 e^{-t^3}$ as a pdf of a Weibull random variable directly as well, which is constructed by taking an exponential random variable X and taking it to the power $\frac{1}{3}$. This is done implicitly in our generation method.